

Antigravity Document Operating System

Comprehensive User Guide & Practical Operational Manual

VERSION 3.0

The Engineering Standard for Reliable AI Document Operations

EKPO OTU, Ph.D.

linktr.ee/ekpootu

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CHAPTER 1

Welcome: What is the Document Operating System?

If you have ever asked an AI assistant to "read this PDF", "edit this spreadsheet", or "fix this PowerPoint presentation", you might have noticed that things often go wrong. Standard AI agents frequently hallucinate missing information, accidentally erase vital mathematical formulas in Excel, break document margins, or deliver files that won't even open.

Document OS is your AI agent's reliable co-pilot. Think of it as an automated safety suite, forensic reader, and quality inspector all in one. It gives your Antigravity agent the specialized knowledge and precision tools needed to work with real documents — without making mistakes or deleting your data.

NOVICE FRIENDLY

You do not need to memorize complex terminal commands or write Python code! Simply talk to your Antigravity agent in plain English. Document OS works automatically behind the scenes to safeguard your documents.



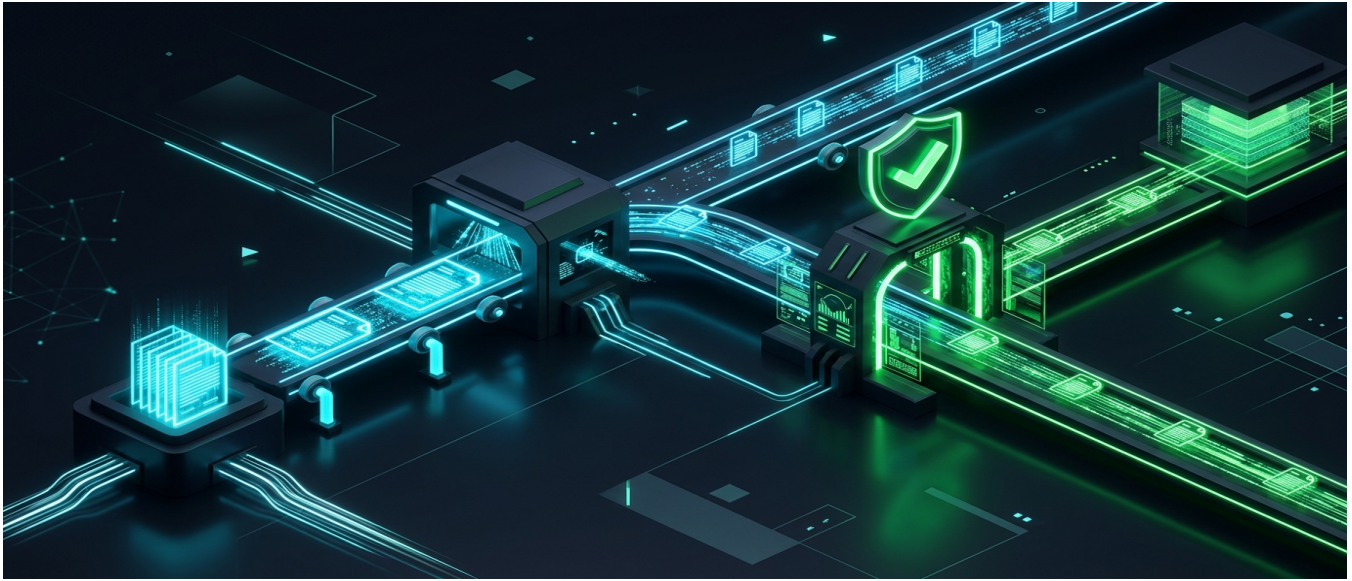
CHAPTER 2

Quickstart: How to Use Document OS Every Day

Using Document OS requires zero technical wizardry. When you converse with your Antigravity agent, the system automatically detects when you are dealing with files. Here are the 5 most common everyday scenarios and how to prompt your agent:

What You Want to Do	What You Prompt	What Document OS Does
Read & Summarize a PDF	"Read the 2024 Annual Report and summarize key revenue figures."	Checks if PDF is digital or scanned, extracts tables cleanly.
Edit an Excel Spreadsheet	"Update Q3 projections with 5% increase. Keep formulas intact."	Loads workbook with formula preservation. Runs automated QA.
Convert Formats Safely	"Convert this Word proposal into a presentation-ready PDF."	Uses headless LibreOffice for 100% fidelity conversion.
Create a Slide Deck	"Turn these notes into a 5-slide PowerPoint in 16:9 layout."	Builds slides with word wrap, applies master templates.
Extract from Scanned Images	"Extract vendor name, date, and total from this receipt."	Applies contrast sharpening, deskewing, then OCR extraction.

The Automated Safety Pipeline



CHAPTER 3

The 5-Stage Safety Pipeline

To ensure zero accidental data loss, every document task passes through a rigorous 5-stage engineering pipeline. Each stage is systematically checked before returning results to you:

Stage	Action	Tool / Helper	Quality Objective
1. Triage	Preflight structural analysis	inspect_doc.py	Determines if PDF is digital or scanned. Counts sheets and formulas. Checks slide aspect ratios.
2. Route	Specialist protocol selection	document-router	Selects the exact safe library (openpyxl with formula-protect, pdfplumber for grids, python-docx for styles).
3. Execute	Deterministic processing	extract_doc.py convert_doc.py	Carries out the operation using pre-tested tools rather than hallucinated code.
4. Render	Visual preview generation	render_doc.py	Converts pages to high-res PNG images so the agent can visually verify layout.
5. QA Audit	Integrity & error check	qa_doc.py	Scans for broken formulas (#REF!, #DIV/0!), confirms files open cleanly.

HOW IT WORKS

The 5-stage pipeline runs automatically — you never need to invoke these tools manually. Your Antigravity agent's Document Router skill triggers each stage in the correct order based on the file type and your request.

CHAPTER 4

Understanding Your Workspace & Global Deployment

Document OS is installed in two complementary locations for total flexibility:

- Local Workspace (./agents/plugins/document-os/): Ready to run in your current project. Commit to Git so your entire team shares the same document rules.
- Global Deployment (~/.gemini/config/plugins/document-os/): Installed into your Antigravity user profile. Every project automatically enjoys Document OS superpowers.

Cross-Harness Freedom

You are never locked into a single AI tool! Document OS adheres to open Agent Skills standards. Run the included exporter to deploy across all your development environments:

```
.\export_cross_harness.ps1 -Target All
```

This bridges your Document OS skills into Claude Code, OpenCode CLI, and Cursor.

CHAPTER 5

Quick Reference CLI Command Cheat Sheet

For power users, automated scripts, or terminal workflows, execute CLI tools directly:

Command	What It Does	Output / Benefit
<code>python inspect_doc.py <file></code>	Analyzes any document format.	Returns JSON with page count, sheet names, formula count.
<code>python extract_doc.py <file> --type tables</code>	Pulls tabular data from files.	Outputs clean Markdown tables for reports.
<code>python convert_doc.py input.docx output.pdf</code>	Headless cross-format conversion.	Vector-perfect without font degradation.
<code>python render_doc.py deck.pptx --outdir ./previews</code>	Renders slides/pages as PNGs.	Vision models can visually QA alignment.
<code>python ocr_doc.py scan.png --output text.txt</code>	Tesseract OCR with image sharpening.	Extracts text from low-contrast scans.
<code>python qa_doc.py model.xlsx</code>	Safety audit for broken formulas.	Asserts 0 errors (#REF!, #DIV/0!) and integrity.

CHAPTER 6

The Zero Unintentional Data Loss Guarantee

Every developer's greatest fear is an AI agent running a destructive command and permanently deleting work. Document OS implements three strict behavioral safeguards:

1. Read-Only by Default: The agent never modifies an original file in place without creating a versioned duplicate (e.g., report_v2.docx) unless you explicitly request it.
2. Formula Protection: If an Excel file contains formulas, the agent is strictly forbidden from replacing them with static numbers.
3. Mandatory Self-Audit: The agent cannot report a task as "Complete" until qa_doc.py validates the resulting file is 100% corruption-free.

SAFETY GUARANTEE

These safety guarantees are encoded as agent rules in GEMINI.md, AGENTS.md, and document-safety.md. They are enforced automatically — you don't need to remember or configure them.

CHAPTER 7

Cross-Harness Freedom: Claude Code, OpenCode, Cursor

Document OS is built on open standards that work across multiple AI agent platforms:

Platform	Installation Path	Export Command
Google Antigravity	.agents/plugins/document-os/	Native — no export needed
Claude Code	~/.claude/skills/document-os/	.\export_cross_harness.ps1 -Target ClaudeCode
OpenCode CLI	~/.config/opencode/skills/	.\export_cross_harness.ps1 -Target OpenCode
Cursor	.cursor/skills/document-os/	.\export_cross_harness.ps1 -Target Cursor

CHAPTER 8

Brand Identity & Professional Document Styling

Document OS v3.0 introduces a built-in Brand Identity Skill that enforces consistent visual standards across all generated documents and web pages. The brand system includes:

- Color Palette: Royal Blue (#1A3A8F), Golden Yellow (#FFC107), Deep Navy (#0D1B4C)
- Typography: Playfair Display for headings, Source Sans 3 for body text, JetBrains Mono for code
- Page Layout: KDP-compliant margins, chapter opener formatting, running headers and footers
- Web Tokens: CSS custom properties for consistent web page styling

The brand tokens are stored in a machine-readable JSON file at `.agents/skills/brand-identity/resources/brand-tokens.json` and can be consumed by any script or template in the Document OS ecosystem.

DOGFOODED

This very guide was generated using the Brand Identity Skill and the professional PDF engine in `build_professional_pdf.py`. Every element — from the chapter openers to the callout boxes to the table styling — follows the brand tokens defined in the skill.

About the Author

Ekpo Otu, Ph.D. is a Computer Professional, Full Stack Developer, Researcher, Author, and Lecturer with a passion for using computational technology to solve real-world problems. His work spans AI and machine learning, agentic systems, document engineering, and open-source tooling.

Platform	Link
Author Bio & Links	linktr.ee/ekpootu
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Document OS is completely free and open source under the MIT License. If this project saved your financial models, corporate decks, or client contracts from being shredded by an AI agent, consider fueling ongoing development!

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